DOCTOR OF PHILOSOPHY

IN

COMPUTER SCIENCE AND APPLICATIONS

OFFERED BY

PANJAB UNIVERSITY, CHANDIGARH
(Examination 2016-17)

Ph.D. Programme

Ph.D. Programme

The purpose of the Doctor of Philosophy programme in computer science is to prepare students for research and teaching positions in universities and colleges, and for research and leadership positions in industry and government. The primary aim of the programme is to train students in the methods of scientific inquiry and independent research. This is accomplished through advanced coursework and active participation with the faculty in their research programmes. Doctor of Philosophy students are expected to have a broad knowledge of all fields of computer science and have a deep understanding of at least one of its areas. In addition to this requirement, a Doctor of Philosophy student must be up to date in all the developments in his/her major area of specialization. The most important component of the Doctor of Philosophy programme is learning to perform independent and significant research in one's area of specialization.

Scheme and Syllabus

First Semester

Sr. No.	Course No.	Course Title	L	T	P	Credits
1	CSA - 101	*Scientific Research Methodology*	**3**	**-**	**1**	**3**
2	CSA - 102	*Research Seminar*	**2**	**-**	**-**	**2**
3	**Elective I:** CSA -103 *Advanced Software Engineering* CSA -104 *Network Technologies* CSA -105 *Advanced Artificial Intelligence and Image Processing Techniques* CSA- 106 *Parallel and Distributed Computing* CSA -107 *Educational Technology and Techniques*		**3**	**-**	**1**	**3**

Scientific Research Methodology

Paper Code: **CSA 101**
L T P : 3 0 1
Objectives: The main objective of the course is to provide a deep and systematic understanding of processes involved in conducting computer science research. The course also enables the students to undertake independent research and prepare thesis.

SECTION–A

Introduction: Meaning of Research, Objectives of research, Types of research, Research Process, Criteria of Good Research

Defining the Research Problem: Need of defining the problem, selecting the problem, techniques involved in defining a problem

Research Design: Need for Research design, features of a good design, different Research designs, Principles of experimental designs, developing a Research plan, steps in sampling design, types of sample design, selecting random sample

Methods of Data Collection: Collection of Primary and secondary data, collection of data through observation, interviews, questionnaires and schedules

Processing and Analysis of Data: Issues in processing, types of analysis, statistics in Research - central tendency, Dispersion, Skewness, Regression analysis, Multiple correlation and regression, partial correlation.

SECTION – B

Sampling Fundamentals: Need, definitions, Central Limit Theorem, Sandler's A- test**,** Standard error, sample size and its determination, Measurement and Scaling Techniques.

Testing of Hypothesis: Hypothesis testing based on Mean, Variance, Correlation Coefficients; limitations of the tests, Chi-square Test, conditions for the application of the test, steps in applying the test, important characteristics of the test, issues in using the test, important nonparametric tests

Analysis of Variance and Covariance (ANOVA): Basics, ANOVA technique, one-way ANOVA, two-way ANOVA, Analysis of Co-variance (ANOCOVA) technique, Multivariate analysis techniques

Interpretation and Report Writing: Technique of interpretation, significance of report writing, steps in writing Research report, types of reports, Oral presentation

Using Statistical Tools: Features of various statistical tools such as MS Excel, SPSS, MATHEMATICA, and MATLAB; application of these tools for analysis of research work

Books:

1. C. R. Kothari : Research Methodology – Methods and techniques (Second Revised Edition) New Age International Publishers 2009 (Chapter 1 to Chapter 9)
2. R. pannerselvam : Research Methodology, PHI, 2005
3. Paul Connolly : Quantitative Data Analysis in Education: A Critical Introduction using SPSS
4. Kiran Singh, B.B. Chaudhuri : MATLAB Programming, PHI
5. James Gips : Mastering MS Excel 2007: A Problem Solving Approach
6. Amos Gilat : MATLAB: An Introduction with Applications, 3rd Edition, 2008, Wiley

Research Seminar

Paper Code: **CSA 102**
L T P : 2 0 0

This course is a seminar course for Research Scholars in Computer Science and Applications. The aim is to explore recent papers of interest from reputed and referred Journals in Computer Science. Each student should prepare and present one paper as well as write a report on the paper.

After the completion of this course, the student shall be able to

- Demonstrate and develop fundamental research skills such as literature search, hypothesis/question formation, presentation of findings and critical assessment of conclusions and implications
- Specify and design a major research project

Advanced Software Engineering

Paper Code: **CSA 103**
L T P : 3 0 1
Objectives: This comprehensive course provides a broad perspective on the advanced topics of Software Engineering and embark on more in-depth research or practice in software engineering.

SECTION–A

Software Architecture - concepts, representation techniques, development methods, and tools for architecture-centric software engineering, architecture description languages, decentralized software architecture, evaluating software architectures.

Pattern-Oriented Software Architecture, Distributed Software Architectures using Middleware, Enterprise Application Integration, Architectures for Mobile and Pervasive Systems, Model Driven Architecture, Component based software developments

SECTION – B

Quality: Quality issues in the life-cycle model. Quality planning and management. Aspects of quality: reliability, maintainability, correctness, usability. Risk analysis and management.

Measurement: The fundamentals of Measurement Theory. Introduction to Metrics (Quality metrics, cost metrics and process metrics). Function points and object points. Estimation Models and Techniques (Top-down/Bottom Up, Analytical Models (Cocomo, Putnam, etc.), Custom Models).

Testing: Requirements, Unit, integration and system testing, validation. Structural testing: coverage techniques. Behavioural testing: domain testing, finite state testing. Mutation and fault seeding. Tools and instrumentation.

Books:

1. Mary Shaw and David Garlan. : Software Architecture: Perspective on an Emerging Discipline, Prentice –Hall.
2. Paul Clements, Rick Kazman, and Mark Klein. : Evaluating Software Architecture Methods and Case Studies. Addison Wesley.
3. Len Base, Paul Clements, Rick Kazman. : Software Architecture in Practice. Second Edition, Addison Wesley.
4. I. Sommerville : Software Engineering, Addison Wesley.
5. Erich Gamma, Richard Helm, Ralph Johnson, and John Vlissides : Design Patterns, Addison-Wesley
6. Stephen T. Albin : The Art of Software Architecture: Design methods and Techniques, Wiley
7. David Budgen : Software Design, 2nd edition, Addison-Wesley
8. William A. Florac and Anita D. Carleton Measuring the Software Process; Addison-Wesley
9. Fenton, Norman E., Whitty & Iizuka Software Quality Assurance & Measurement: A Worldwide Perspective, Thomson Computer Press
10. Horch J. W Practical Guide to Software Quality Management, Artech House Computer Science
11. Kan Stephen H. Metrics and Models in Software Quality Engineering, Addison-Wesley.
12. Glenford J. Myers The Art of Software Testing, John Wiley & Sons

Network Technologies

Paper Code: **CSA 104**
L T P : 3 0 1
Objectives: This challenging and comprehensive course provides a broad perspective on the wireless networks.

SECTION–A

Overview of wireless networks: Introduction to wireless communication, IEEE 802.11, Wireless ATM, IEEE 802.16 and IEEE 802.20. Review of GSM

IPv6: Basic protocol, extensions and options, support for QoS, security, neighbor discovery, auto-configuration, routing.

WiMAX Networks: Architecture, MAC layer, physical layer, spectrum allocation issues, comparison with WiFi and limitations.

Traffic Management: Economic Framework, Traffic Models, Traffic Classes, Scheduling, Admission Control, Peak Load Pricing

QoS and Security issues: Network security requirements, issues and challenges in security and QoS provisioning, classifications of QoS solutions, IETF integrated services model, Differentiated Services Model. Flow identification, Packet Classifiers and Filters.

SECTION – B

Mobility in networks: Mobile IP and related issues like Route Optimization, Handoff, and Security.

Mobile Ad hoc Networks (MANETs): Introduction to Ad hoc wireless networks and sensor networks, Various MANET issues.

Transport layer: Various transport layer solutions and TCP over Ad hoc wireless networks, TCP extensions for high-speed networks, transaction-oriented applications, other new options in TCP, TCP in Wireless Domain

Network Simulation: Elements of queuing model, role of exponential distribution, Event graphs of queuing model, Discrete-event and Continuous Simulation, Introduction to NS2 and Qualnet.

Books:

1. William Stallings : Wireless Communication and Networks, Pearson
2. C. Siva Ram Murthy and B. S Manoj : Adhoc Wireless Networks-Architecture and Protocols, Pearson Education
3. W. R. Stevens : TCP/IP Illustrated, Vol. 1, Pearson Education
4. M. Gonsalves and K. Niles : IPv6 Networks, McGraw Hill
5. S. Keshav : An Engineering Approach to Computer Networking, Pearson Education
6. Hamdy A. Taha : Operations Research and Introduction, Pearson Education
7. Requests for Comments (RFCs) & Internet Drafts, published by Internet Engineering Task Force (www.rfc-editor.org)
8. Journals: IEEE Journal on Selected Areas in Communications, IEEE Transactions on Communication, ACM/IEEE Transactions on Networking

Advanced Artificial Intelligence and Image Processing Techniques

Paper Code: **CSA 105**

L T P : 3 0 1

Objectives: The main objective of the course is to provide understanding of computational techniques useful for research in Artificial Intelligence and Image Processing. After going through the course the researcher will be able to choose and apply the techniques required for the research problem undertaken.

SECTION–A

Fuzzy Logic: Introduction, Basic Concepts of Fuzzy Logic, Fuzzy Sets, Fuzzy Relations, Fuzzy Graphs, and Fuzzy Arithmetic, Fuzzy If-Then Rules, Fuzzy Implications and Approximate Reasoning, Fuzzy Logic in Control Engineering, Fuzzy Logic and Artificial Intelligence, Fuzzy Logic in Database Management and Information Systems, Fuzzy Logic in Pattern Recognition, Neuro-Fuzzy Systems, Genetic Algorithms and Fuzzy Logic.

Computational Intelligence: Introduction, Basics of Computational Intelligence, Evolutionary Computation, Machine Learning, Parallel and distributed Artificial Intelligence, Agent and Multi-Agent systems,

Genetic Algorithms: Fundamentals, Encoding, Fitness function, Reproduction, Inheritance Operators, Crossover, Inversion and Deletion, Mutation Operator, and Bit-wise Operators; applications

Bayesian Networks: Reasoning under uncertainty**,** Bayesian Networks, Inference in Bayesian Networks, Decision Networks, Applications of Bayesian Networks.

SECTION – B

Pattern Recognition and Image processing: Pattern Recognition Process, Representation, Digital Image processing, Image Transformation, Image enhancement and Image Analysis: Edge Detection, Segmentation, Feature extraction and Classification.

Neural Networks: Introduction to Artificial Neural Network, Neural Networks Architectures, Supervised and Unsupervised Learning, Preceptron, Back Propagation Networks and Associative Memory.

Data Warehousing and Data Mining: Building Data Warehouses, Data Warehouse Architectural Strategies, Tools for Data Warehousing, Data parts and its types, Applications. Data Mining**:** From Data Warehouse to Data-Mining, Steps of Data Mining Process, Tools for Data – Mining, Data Mining Applications.

Programming using Tools such as MATLAB, GeNIe, JADE (Java Agent Development Environment) and Java: Features and Applications

Books:

1. Reza Langari, John Yen : Fuzzy Logic: Intelligence, Control, and Information, 1st Edition, Pearson (Chapters 1-8, 11-13, 16, 17)
2. Laurene V. Fausett : Fundamentals of Neural Networks: Architectures, Algorithms And Applications, 1993, Prentice Hall. (Chapters 1-6)
3. Timothy Ross : Fuzzy Logic with Engineering Applications, 3rd Edition 2010, Wiley
4. Kevin B Korb, Ann Nicholson, Korb B Korb : Bayesian Artificial Intelligence, 2004, Chapman & Hall/CRC. (Chapters 1-5)
5. Russell C. Eberhart, Yuhui Shi : Computational Intelligence: Concepts to Implementation, 1st Edition, 2007. Morgan Kaufman
6. David hand, Heikki Mannila, Padhraic Smyth : Principles of Data Mining, PHI
7. S Rajasekharan, G.A. Vijayalakshmi Pai Neural Networks, Fuzzy Logic and Genetic Agorithms, PHI
8. Gerhard Weiss Multiagent Systems edited by, MIT press 1999.
9. Amos Gilat Amos Gilat, MATLAB: An Introduction with Applications, 3rd Edition, 2008, Wiley
10. Rafael C. Gonzalez and Richard E. Woods Digital Image Processing

Parallel and Distributed Computing

Paper Code: **CSA 106**
L T P : 3 0 1
Objectives: This comprehensive course provides a broad perspective on the advanced topics of Parallel and Distributed Computing.

SECTION–A

Parallel Computing: Taxonomy of parallel processing. Parallel techniques and paradigms; Abstract models of parallel machines; Strategies and paradigms for parallel algorithm design; Correlations of algorithmic to architectural spaces;

Design methodologies for parallel algorithms for various parallel computer platforms; Designing parallel algorithms for various application areas: linear algebra, matrix multiplication, sorting, PDE, graphs, combinatorial search, document classification;

Design methodologies for parallel genetic algorithms for various parallel computer platforms. Parallel programming with MPI, Parallel programming with OpenMP, Hybrid parallel programming MPI+OpenMP

Multithreaded programming, Parallel program benchmarking, Parallelism profiling

SECTION – B

Cluster Computing: The Role of Clusters, Architecture and Cluster Software, Network Hardware, Network Software, Protocols, Distributed File Systems, Virtualization technologies, Benchmarks

Grid Computing: Role of a Grid, infrastructure**,** Grid Architecture, Overview of Resource Managers, Application Management; Grid Application Description Languages, Application Partitioning, Meta-scheduling, Mapping ,Monitoring, Web Services, Grid Portals

Books:

1. Grama A., Gupta A., Karypis G., Kumar V : Introduction to Parallel Computing, PEARSON, Addison Wesley
2. Barry Wilkinson, Michael Allen : Parallel Programming Techniques and Applications Using Networked Workstations and Parallel Computers Pearson Prentice Hall .
3. Haupt R., S. Haupt : Practical Genetic Algorithms, Wiley Interscience, A John Wiley & Sons
4. M. Quinn : Parallel Programming in C with MPI and OpenMP, McGraw Hill
5. Hwang K., Zhiwei X. : Scalable Parallel Computing: Technology, Architecture, Programming, McGraw-Hill.

Educational Technology and Techniques

Paper Code: **CSA 107**
L T P : 3 0 1
Objectives: Objectives: The main objective of the course is to provide a deep and systematic understanding of processes involved in conducting research in E-learning and Instructional Design techniques .

SECTION - A

Learning and E-learning Connect: Genesis , evolution**;** educational technology and interaction
Panacea called E-learning : Components, architecture, Models; characterstics of e-learning and e-Knowledge, Dynamic content generation and management: services , technology**;**
Learning Design : critical perspective and use of design patterns for e-learning , developing taxanomy for learning designs, learning design representations to document, model and share teaching practice

The instructional Design process: General guidelines for Instruction Design; Basic content research methodologies; Designing instructions for Classroom and online learning.
Learning Objects and Generative learning for higher order thinking: design of Learning Object for pedagogical impact, Tools , techniques and technology in Open Source for Effective usage of Learning Object ,Learning Management Systems, Learning Content Management Systems, Learning Object repositories

Standards for Learning Objects and learning designs: SCORM, IMS etc**;** Principled construction and reuse of Learning Objects using standards**;** Design, development and evaluation of applications using the techniques.

SECTION – B

Simulation & Modeling: Fundamental concepts, types of simulation, types of models, continuous and discreet system simulation, static and dynamic simulation, techniques of simulation, design of simulation experiments, validation of simulated model, Simulation and M odeling tools
Visualization: Fundamentals, techniques and applications
Finite Element Analysis: Finite element method and practical application to different problems
E-content development: Use of commercial/open source software for developing e- learning material

Books:

1. Lori Lockyer, Sue Bennett : Learning Design and Learning Objects Issues, Applications and Technologies VolumeII, Information Science Press
2. Marc-Alain Steinemann : Distributed Architectures for Laboratory based E-learning Llumina Press
3. Ruth Colvin Clark & Richard E. Mayer : E-learning and the Science of Instruction Second Edition. Pfeiffer. A Wiley Imprint
4. William Horton : -Learning by design Pfeiffer. A Wiley Imprint
5. Som Naidu : Learning and Teaching with Technology Learning and Teaching with Technology Routledge Taylor & Francis Publication
6. Averill, M. Law, W. David Kelton, : Simulation Modeling and Analysis, 3/e, Mc Graw Hill publications, 2000
7. C. S. Krishnamoorthy, : Finite Element Analysis; Theory and Programming
8. E-learning concepts and techniques : http://iit.bloomu.edu/Spring2006_eBook_files/
9. Willis, Jerry, : Qualitative Research Methods in Education and Educational Technology